

Akshat S. Chaturvedi

Curriculum Vitae

Georgia State University Department of Physics and Astronomy

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EDUCATION

Georgia State University

PhD Astronomy (expected)

Atlanta, GA

2024 – Present

Georgia State University

MS Physics w/ Astronomy Concentration

Atlanta, GA

2024 – 2026

Pennsylvania State University

BS Astronomy & Astrophysics; BS Physics; Minor in Mathematics

University Park, PA

2020 – 2024

RESEARCH EXPERIENCE

Georgia State University

Graduate Research Assistant

Atlanta, GA

Fall 2024 – Present

Advisor: Dr. Douglas R. Gies

Project: *Investigating the fundamental properties of Be+sdOB binaries*

Research Expertise: Be stars, UV and optical spectroscopy, radial velocity analysis, cross correlation function analysis, Doppler tomography, long-baseline optical/near-IR interferometry

Pennsylvania State University

Undergraduate Research Assistant

University Park, PA

Fall 2023 – Summer 2024

Advisors: Dr. Robin Ciardullo and Dr. Howard E. Bond

Project: *A spectroscopic study of the peculiar planetary nebula Abell 57*

Research Expertise: Planetary nebulae, optical spectroscopy, photoionization modeling, UV photometry

Pennsylvania State University

Undergraduate Research Assistant

University Park, PA

Fall 2022 – Present

Advisors: Dr. Howard E. Bond, Dr. Robin Ciardullo, and Dr. Gautam Nagaraj

Project: *A census of blue post-AGB stars in Galactic globular clusters*

Research Expertise: Post-AGB stars, globular clusters, optical photometry, clustering algorithms, stellar populations, astrometric analysis

Pennsylvania State University

Undergraduate Research Assistant

University Park, PA

Summer 2022

Advisors: Dr. Kevin Luhman

Project: *A chrono-kinematic analysis of nearby star associations*

Research Expertise: Clustering algorithms, stellar populations, kinematic analysis, traceback age analysis

PUBLICATIONS

Detection and Astrometry of the Ba–Bb Subsystem in α Piscium: First Dual-field Interferometry at the CHARA Array

N. Anugu, R. Klement, [...] **A. S. Chaturvedi**, et al. 2026

The Astronomical Journal, Volume 171, Issue 4, id. 253

Broad-band Monitoring of 797 Montana

M. C. Bentz, R. Alakhone, L. Azoulay, E. Burns-Kaurin, **A. S. Chaturvedi**, et al., 2025

The Minor Planet Bulletin Vol. 52, No. 3, pp. 207-208

R-Band Monitoring of Patroclus and Menoetius Mutual Events

M. C. Bentz, R. Brown, S. Carrazco-Gaxiola, **A. S. Chaturvedi**, et al., 2025

The Minor Planet Bulletin Vol. 52, No. 1, pp. 79-81

Spectroscopic Survey of Faint Planetary-nebula Nuclei. V. The EGB 6-type Central Star of Abell 57

H. E. Bond, **A. S. Chaturvedi**, et al., 2024

The Astrophysical Journal, Volume 970, Issue 2, id.164

CONFERENCE POSTERS & TALKS

2026 CHARA/MROI Science Meeting

Socorro, NM

Talk: Exploring the Binarity of Be Stars

March 2026

Georgia Regional Astronomy Meeting

Atlanta, GA

Talk: Be Stars and Beyond: A Legacy of Binary Mass Transfer

November 2025

Staraganza III

Atlanta, GA

Talk: Be Stars and Beyond: A Legacy of Binary Mass Transfer

September 2025

Galaxies to Gluons Summer Lunch Talk

Atlanta, GA

Talk: Be Stars and Beyond: A Legacy of Binary Mass Transfer

July 2025

Penn State Undergraduate Research Exhibition

University Park, PA

Poster: A Spectroscopic Study of the Peculiar Planetary Nebula Abell 57

April 2024

American Physical Society Mid-Atlantic Section Meeting

University Park, PA

Poster: A Census of Blue Post-Horizontal-Branch Stars in Galactic Globular Clusters

Nov 2025

AWARDS, FUNDING, & SCHOLARSHIPS

Second Century Initiative (2CI) Fellow

Fall 2024 – Present

Georgia State University

Faculty Undergraduate Research Support Award

Spring 2024

\$1000 to support publication costs

Pennsylvania State University

Induction into Sigma Pi Sigma <i>National physics and astronomy honors society</i> Pennsylvania State University	Spring 2024
Induction into Phi Beta Kappa <i>National liberal arts and sciences honors society</i> Pennsylvania State University	Spring 2023
Eberly College of Science Dean's List Pennsylvania State University	2021, 2022, Fall 2023, and Spring 2024
George E., Jr. and Elizabeth S. Sperling Tutorial Endowment Scholarship <i>\$1000 awarded to learning assistants who excel in tutorial ability</i> Pennsylvania State University	Spring 2022

OUTREACH & LEADERSHIP

Georgia State University Undergraduate Research Conference Graduate Student Judge for Physics and Astronomy Posters	Atlanta, GA April 2026
Astronomy on Tap Atlanta Talk: Final Destination: The Epic Ends of Stellar Lives Viewed by Next Generation Telescopes	Atlanta, GA August 2025
AsTropaLooza Citizen Science Workshop – Volunteer Astronomy of Atlanta K–12 STEM Outreach Event	Atlanta, GA June 2025
Department of Physics & Astronomy Observing Nights – Volunteer Observing Nights for Astronomy Students	Atlanta, GA Spring 2025 – Present
Hard Labor Creek Observatory Public Open Nights – Volunteer Public Open House at Georgia State University's Hard Labor Creek Observatory	Rutledge, GA Fall 2024 – Present
Vice President, Society of Physics Students Pennsylvania State University	University Park, PA Fall 2023 – Spring 2024
Society of Physics Students CV Building Workshop – Lead Pennsylvania State University	University Park, PA Fall 2023

OBSERVING EXPERIENCE

Astrophysical Research Consortium 3.5m Telescope Apache Point Observatory Sunspot, NM — <i>Obtained high-resolution echelle spectra of high- and intermediate-mass stars using the ARCES instrument</i>	December 2025 – Present Observing Time: 24 hours
Center for High Angular Resolution Astronomy (CHARA) Array Mount Wilson Observatory Mount Wilson, CA — <i>Obtained long-baseline near-infrared interferometric observations of Be+sdOB stars</i>	July 2025 – Present Observing Time: 32 hours

Miller 24" Telescope

Hard Labor Creek Observatory
Rutledge, GA

Fall 2024 – Spring 2025
Observing Time: 50 hours

— *Obtained photometric lightcurves of the asteroids 617 Patroclus, Menoetius, 797 Montana, and 4107 Rufino*

Department of Astronomy & Astrophysics 24" Telescope

Pennsylvania State University
University Park, PA

Fall 2022
Observing Time: 20 hours

— *Obtained photometric observations and images of a variety of astronomical objects including planetary nebulae, globular clusters, and planets*

TEACHING EXPERIENCE

Astronomy 1020K | Stellar and Galactic Astronomy (x9)

Georgia State University
Graduate Teaching Assistant

Fall 2024 – Present
Atlanta, GA

Astronomy 1010K | Astronomy of the Solar System (x1)

Georgia State University
Graduate Teaching Assistant

Spring 2025 – Present
Atlanta, GA

Astronomy & Astrophysics 292 | Astronomy of the Distant Universe

Pennsylvania State University
Undergraduate Learning Assistant

Spring 2023
University Park, PA

Astronomy & Astrophysics 291 | Astronomical Methods and the Solar System

Pennsylvania State University
Undergraduate Learning Assistant

Fall 2022
University Park, PA

Physics 212M | General Physics: Electricity and Magnetism

Pennsylvania State University
Undergraduate Learning Assistant

Spring 2022
University Park, PA

Physics 212M | General Physics: Mechanics

Pennsylvania State University
Undergraduate Learning Assistant

Fall 2021
University Park, PA

SOFTWARE

CATCH (CHARA Array's Thrifty Calibrator Hunter)

A tool to find and verify calibrator stars for interferometry targets at the CHARA Array

TECHNICAL SKILLS & LANGUAGES SPOKEN

Programming: Python, ADQL, R, Julia, Rust

Software: TOPCAT, CLOUDY, AstroImageJ, SAO DS9, IRAF

Languages: English (native), Hindi (native), Marathi (native), French (A2)